



Data input/output methods & Advanced computer architecture

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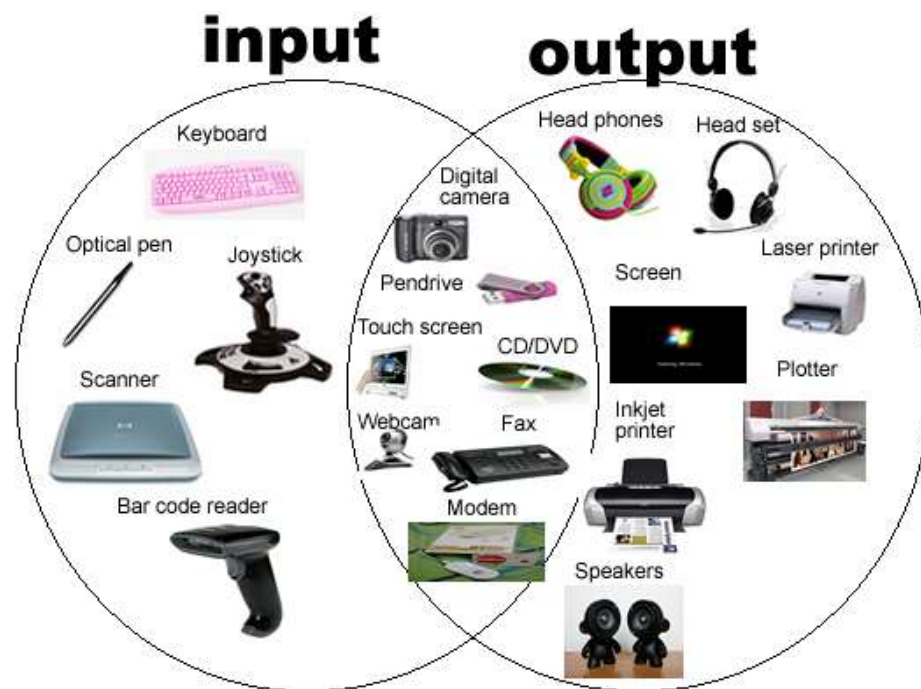
- Introduction to data input/output methods
- Polling input/output
- Interrupts and interrupt handling
 - Interrupts and interrupt classification
 - Interrupt handling cycle
 - Interrupt input/output
- DMA input/output
- Microprocessors and advanced technologies

❖ Role of input/output:

- It is the means by which the CPU communicates with the outside world
- It provides input data for the CPU to process
- It provides a means for the CPU to output data

❖ Main input/output methods:

- Polling
- Interrupt
- Direct Memory Access (DMA)



❖ Computer input/output ports:

- PS/2: port for connecting to keyboard and mouse
- COM: serial ports
- LPT: parallel ports
- IDE, SATA, SCSI: ports for connecting drives
- LAN: port for connecting to local area network
- Audio: ports for connecting audio (speaker, mic, and line-in)
- Video: ports for connecting to monitors (analog)
- DVI: ports for connecting to monitors (digital)
- USB: ports for connecting according to USB standards: USB 1.0: 12Mb/s; USB 2.0: 480Mb/s; USB 3.0: 1.5Gb/s

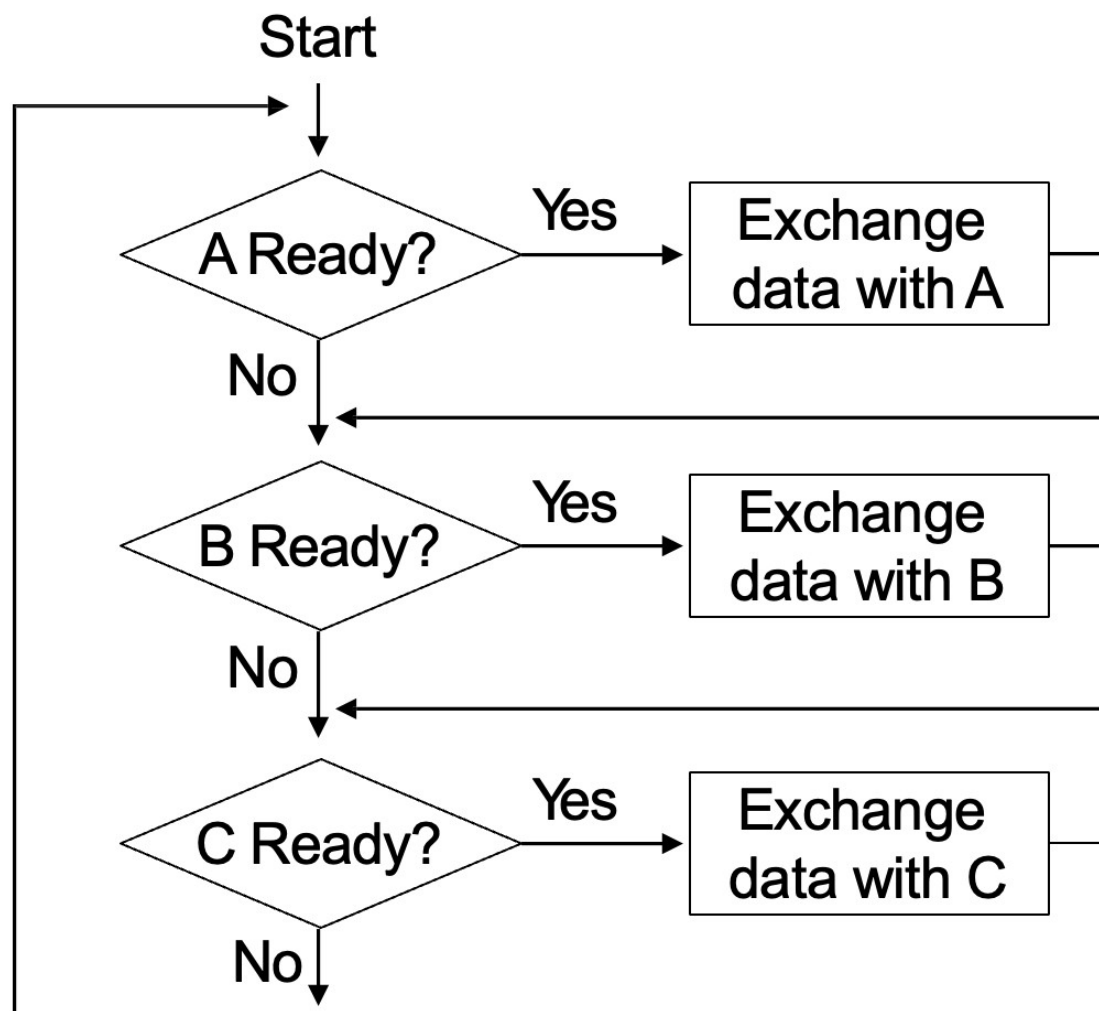
❖ Polling-based input/output mechanism:

- The CPU manages a list of input/output devices along with their communication port addresses;
- Input/output devices periodically update their readiness status using their input/output status flag bits;
- The CPU periodically scans the input/output devices to read their status flag bits;
 - If a device is ready for operation, the two sides exchange data;
 - After data exchange is complete, the CPU continues scanning for other devices.
- The CPU is the proactive party in the data exchange process.

❖ Advantages and disadvantages of polling I/O:

- Advantages:
 - Simple, easy to implement
 - Can be implemented using software
- Disadvantages:
 - Low efficiency because the CPU spends a lot of time polling devices
 - Not really feasible when there are many devices in the polling list
- Applications of polling I/O:
 - Polling is often used during system startup: The CPU polls most devices to establish configuration
 - Polling is used during operation with removable devices such as CD/DVD drives, floppy drives, etc.

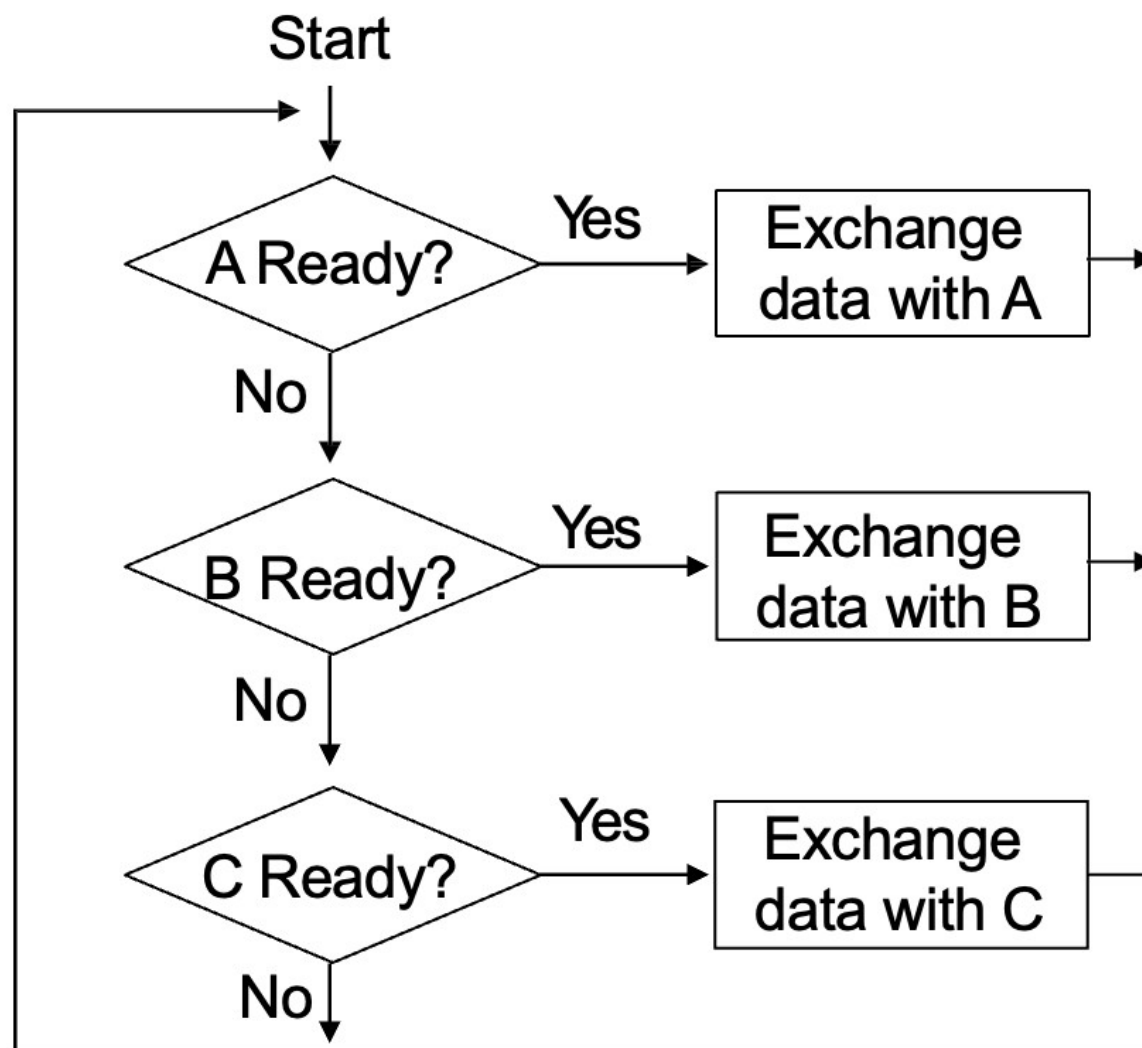
❖ Polling-based input/output:



❖ Polling-based input/output:

- Three devices A, B, and C are polled without priority.
 - The CPU scans all devices in a poll cycle.
 - The CPU can exchange data with more than one device in a poll cycle.
 - Devices are “visited” sequentially, regardless of the device preceding the cycle.
 - The CPU starts a new poll cycle after scanning all devices.

❖ Characteristics



❖ Polling-based input/output - prioritized:

- Three devices A, B, and C are polled with priority in the order: A, B, C
 - The CPU scans all devices in a poll cycle;
 - The CPU exchanges data with a maximum of 1 device per poll cycle;
 - High-priority devices are always visited first;
 - Low-priority devices are only visited if the devices before them are unavailable.
 - The CPU starts a new poll cycle immediately after exchanging data with a device.

❖ What is an interrupt?

- An interrupt is an event where the CPU temporarily pauses the execution of a program to execute another program segment at the request of an external party;
- Typically, these external requests originate from input/output devices. These requests are called interrupt requests;
- The program segment the CPU executes during the interrupt is called an interrupt service routine (ISP).

- ❖ ISPs are program segments:
 - Pre-written and stored in ROM;
 - Each ISP has its own task and is usually responsible for exchanging data with input/output devices.
- ❖ When does the CPU check and handle interrupts?
 - The CPU checks for interrupt requests at the last clock cycle of the instruction cycle.

❖ Interrupt Classification:

- Hardware Interrupts: These are interrupts triggered by hardware components sent to the CPU's NMI and INTR pins, including:
 - Non-Maskable Interrupts: interrupts sent to the CPU's NMI pin, not affected by interrupt flags; e.g., Reset interrupt;
 - Maskable Interrupts: interrupts sent to the CPU's INTR pin, affected by interrupt flags; IF=1 -> enables interrupt, IF=0 -> disables interrupt.
- Software Interrupts: These are interrupts triggered by programs through the interrupt call instruction `INT<N>`. N is the interrupt number, N = 0-255.
- Exception Interrupts: These are interrupts caused by errors that occur during CPU operation:
 - Divide by Zero Interrupt
 - Overflow Interrupt

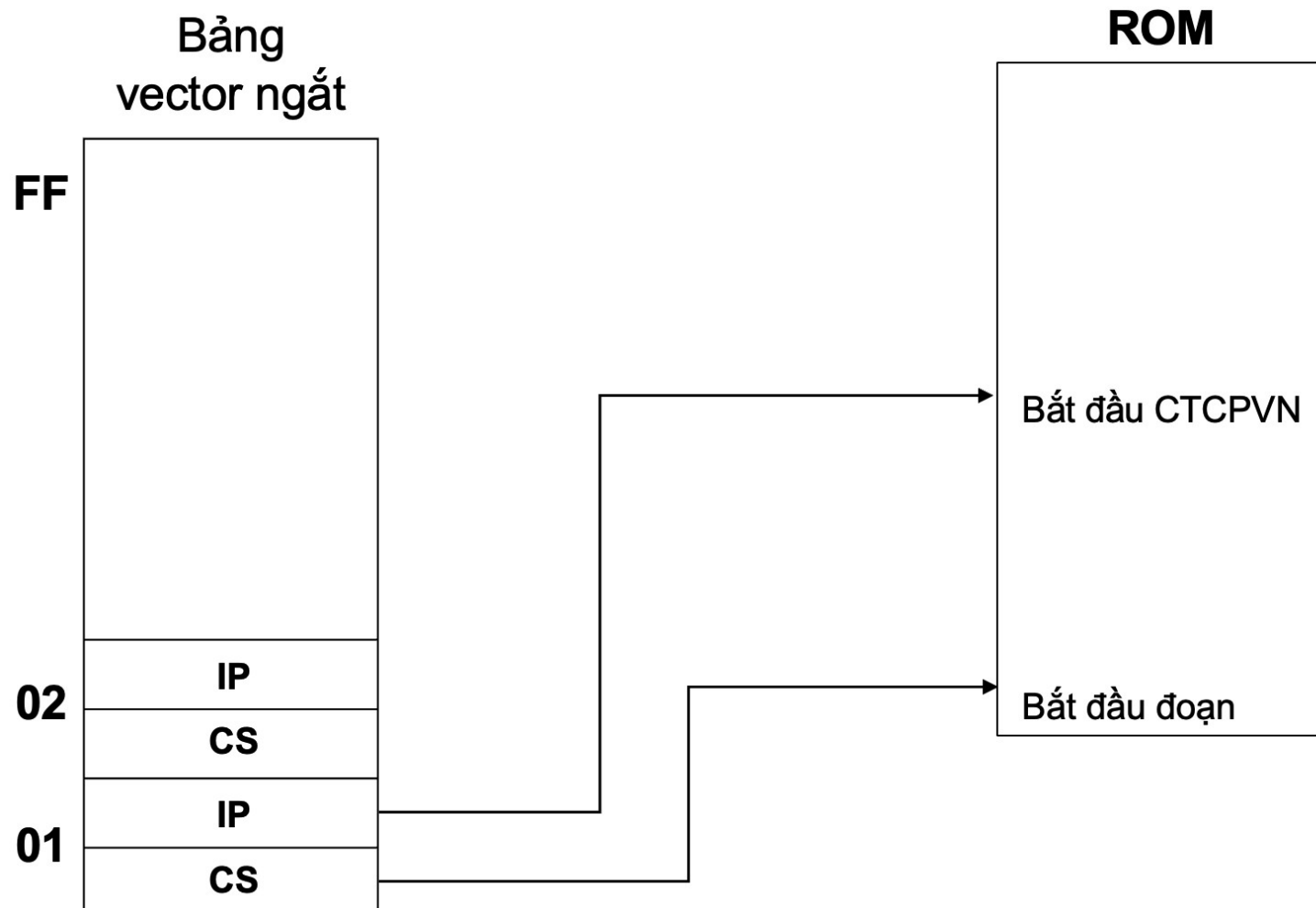
❖ Interrupt Request Priority Order

- Interrupt requests are assigned a priority level
- When multiple interrupt requests are received simultaneously, the CPU will process them according to the predetermined priority level.
- Interrupt request priority levels (from highest to lowest)
 1. Internal interrupt: INT 0 (divide by zero), INT N ($N \neq 0$)
 2. Non-maskable interrupt NMI
 3. Maskable interrupt INTR
 4. Instruction-by-instruction interrupt: INT 1

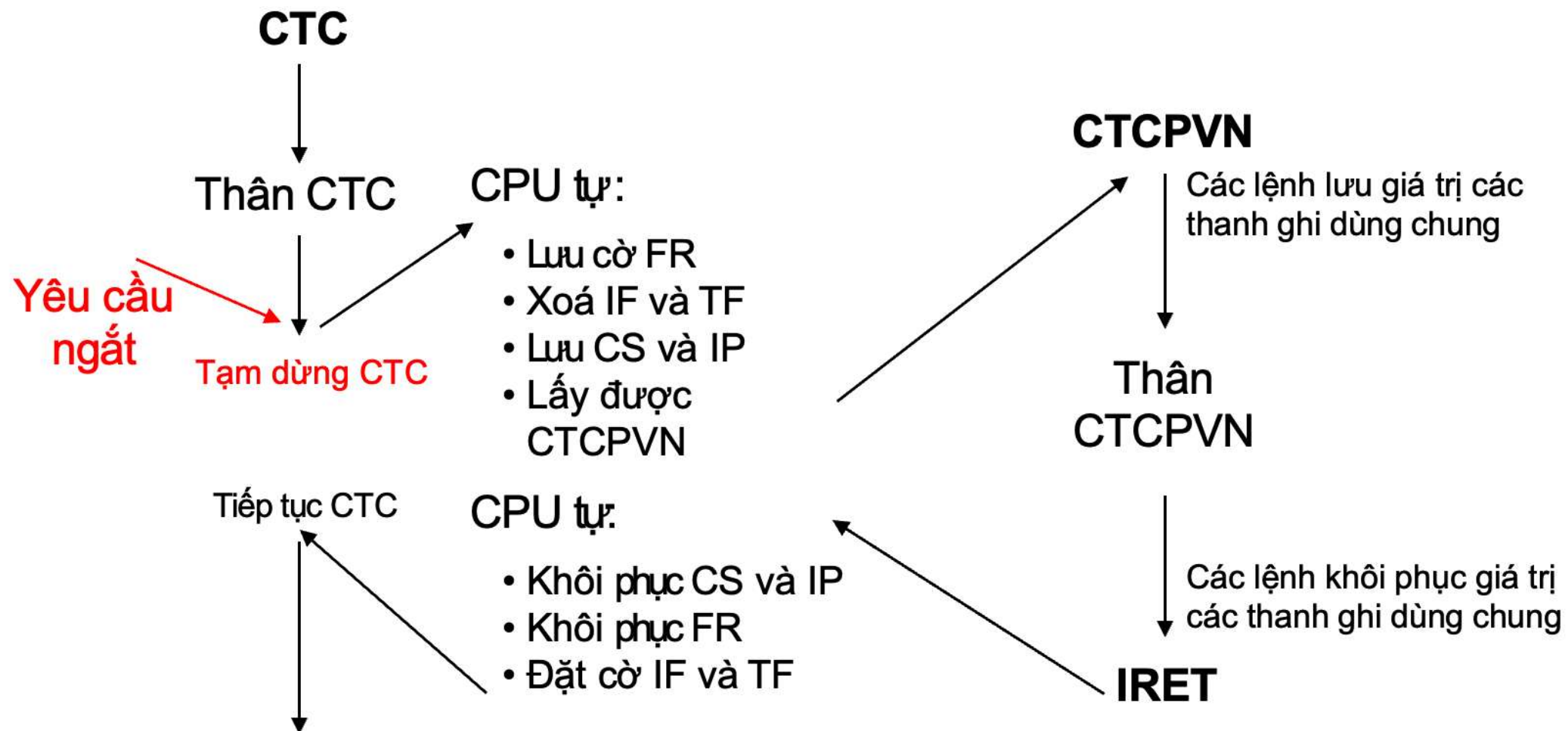
❖ Interrupt Vector Table:

- The 8086/8088 microprocessor has 256 interrupts numbered from 0-255.
- An interrupt vector contains the following information:
 - Interrupt number N, N=0-255 or 00-FFH
 - Full address of the interrupt service routine (SRT) stored in ROM memory. The full address includes:
 - Segment address (CS)
 - Offset address (IP)
- The interrupt vector table stores information about 256 interrupt vectors. Each record in the table contains the following information:
 - Interrupt number
 - Segment address and offset address of the SRT.

❖ Interrupt Vector Table:



❖ Interrupt Handling Cycle:



❖ **Interrupt Handling Cycle:** There are four steps in the interrupt handling cycle as follows:

1. Upon receiving an interrupt request, the CPU performs the following tasks:
 - Completes the current instruction of the main program (CPC)
 - Saves the value of the FR flag register to the stack
 - Clears the IF interrupt flag and TF trap flag
 - Saves the values of the CS and IP registers to the stack
 - From interrupt number N, retrieves the address of the CPC from the interrupt vector table.

❖ **Interrupt Handling Cycle:** There are four steps in the interrupt handling cycle as follows:

2. Loads the address of the CPC into CS and IP, the CPU executes CPC:
 - Saves the values of the shared registers to the stack
 - Executes the main code of the CPC
 - Restores the values of the shared registers.

❖ **Interrupt Handling Cycle:** There are four steps in the interrupt handling cycle as follows:

3. Upon encountering the IRET instruction ending the CPC, the CPU performs the following tasks:
 - Restores the values of CS and IP
 - Restores the value of the FR flag register
 - Sets the IF interrupt flag and TF trap flag.
4. The CPU executes the next instruction of the CPC (following the interrupt instruction).

❖ Interrupt-based input/output - Interrupt-based input/output cycle:

1. An input/output device that needs to exchange data sends an interrupt request to the CPU's INTR signal pin;
2. Upon receiving an interrupt request, the CPU performs the following tasks:
 - Completes the current instruction of the main program (CPC)
 - Saves the value of the FR flag register to the stack
 - Clears the IF interrupt flag and TF trap flag
 - Saves the values of the CS and IP registers to the stack
 - Sends an interrupt acknowledgment signal to the input/output device via the INTA signal pin.

- ❖ Interrupt-based input/output - Interrupt-based input/output cycle:
3. Upon receiving the CPU's interrupt acknowledgment signal, the input/output device sends interrupt number N to the CPU.
 4. Upon receiving interrupt number N, the CPU retrieves the address of the corresponding CPVN from the interrupt vector table.
 5. Loading the CPVN address into CS and IP, the CPU executes CPVN, which includes:
 - Saving the values of the shared registers to the stack
 - Executing the main code of the CPVN: simultaneously exchanging data with the input/output device
 - Restoring the values of the shared registers.

❖ Interrupt-based input/output - Interrupt-based input/output cycle:

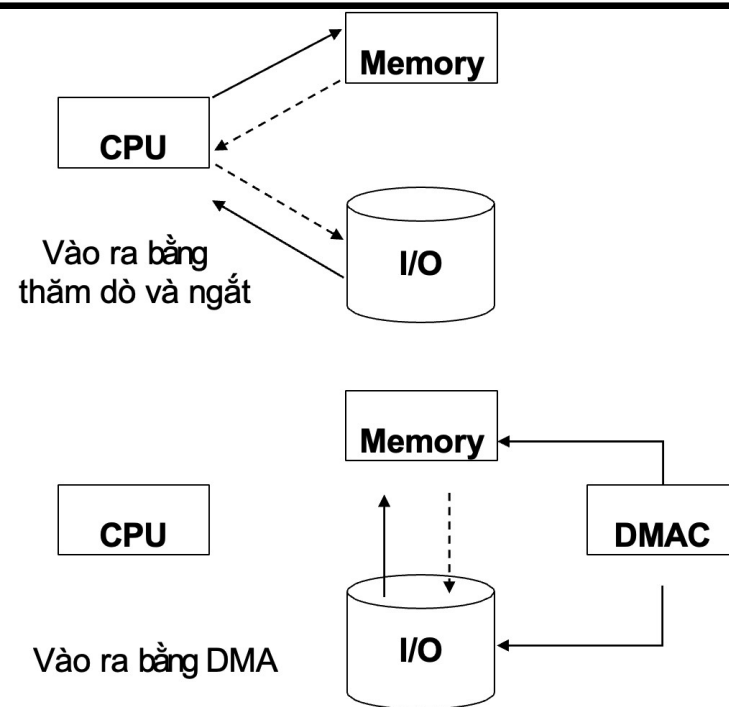
6. Upon encountering the IRET instruction ending CPVN, the CPU performs the following tasks:
 - Restores the values of CS and IP
 - Restores the value of the flag register FR
 - Set the IF interrupt flag and TF trap flag
7. The CPU continues executing the next instruction of the CTC (immediately following the instruction where the interrupt occurred).

❖ Advantages and Disadvantages of Interrupt-Based I/O

- Advantages:
 - More efficient than polling I/O, as the CPU does not have to poll each device
- Disadvantages:
 - More complex than polling I/O
 - Requires hardware circuitry to control interrupts
- Active party in interrupt-based I/O:
 - I/O device

❖ Introduction to DMA (Direct Memory Access) Input/Output:

- In polling and interrupt input/output methods, the input/output device exchanges data with memory via the CPU;
- DMA allows the input/output device to exchange data directly with memory in blocks without going through the CPU;
- DMA is suitable when large amounts of data need to be exchanged in a short period of time.

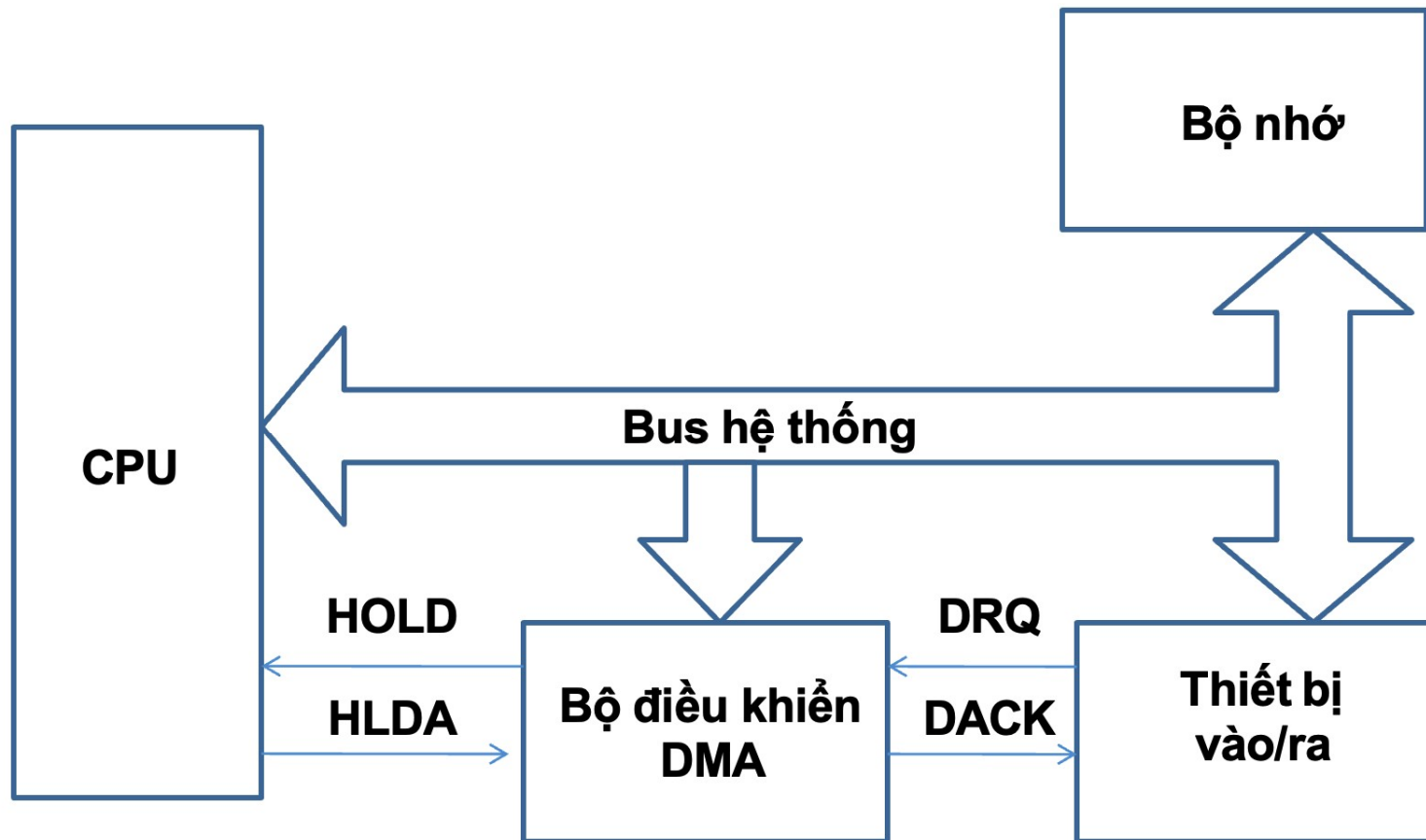


❖ DMA Input/Output:

- The DMAC (DMA Controller) controls the direct data exchange between the input/output device and memory on behalf of the CPU;
- DMA is many times faster than polling and interrupt input/output.
- For example, with the 8088 microcontroller:
 - DMA input/output takes 4 clock cycles to transfer 1 byte from a peripheral device into memory;
 - CPU input/output takes 39 clock cycles to transfer 1 byte from a peripheral device into memory.

```
1          ; So chu ky dong
2 LAP MOV AL, [SI] ; 10
3      OUT PORT, AL ; 10
4      INC SI      ; 02
5      LOOP LAP    ; 17
6          ; Cong 39 chu ky
```

❖ Microprocessor System with DMAC - DMA Input/Output Cycle:



❖ Microprocessor System with DMAC - DMA Input/Output Cycle:

1. An input/output device requesting data exchange sends a DRQ request to the CPU via the DMAC;
2. The DMAC converts the DRQ request into an HRQ and sends it to the CPU's HOLD signal pin;
3. Upon receiving the HRQ bus usage request, the CPU:
 - Sends data exchange control parameters and a HACK bus usage request acknowledgment signal to the DMAC via the HLDA signal pin;
 - Detaches itself from the system bus (100% of the A and D bus signals and some C bus signals);

❖ Microprocessor System with DMAC - DMA Input/Output Cycle:

4. Upon receiving the HACK, the DMAC takes control of the system bus and sends a DACK acknowledgment signal to the input/output device;
5. The DMAC controls the data exchange process directly between the input/output device and memory;
6. At the end of the DMA process, the DMAC returns control of the bus to the CPU.

❖ Advantages and Disadvantages of DMA

- Advantages:
 - Very high performance because data is exchanged directly in blocks between the input/output device and memory without going through the CPU.
- Disadvantages:
 - More complex than polling and interrupt input/output
 - Requires hardware circuitry to control the DMA process
- Active party in DMA input/output:
 - Input/output device

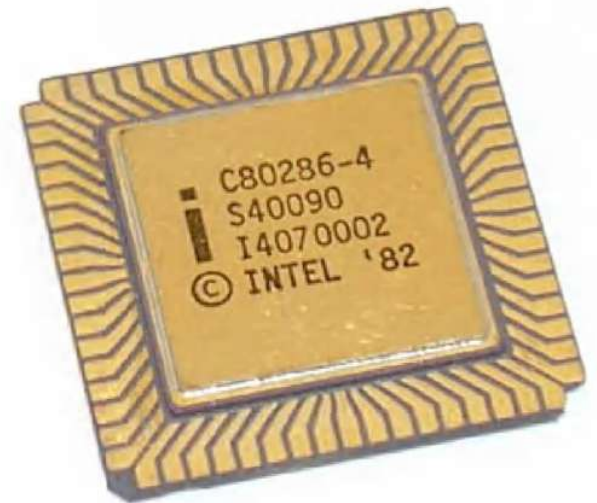
❖ Microprocessors based on IA-32 and Intel 64 architectures.

- x86 microprocessors - 8086
 - The 8086 was the first microprocessor of the IA-32 architecture
 - 16-bit registers
 - Bus: 16-bit data bus; 20-bit address bus.
 - Maximum memory space: 1MB
 - Memory is segmented: each segment is 64KB in size.
 - Logical memory address:
 - Segment: Offset
 - Segment address: Offset address.



❖ Microprocessors based on IA-32 and Intel 64 architectures

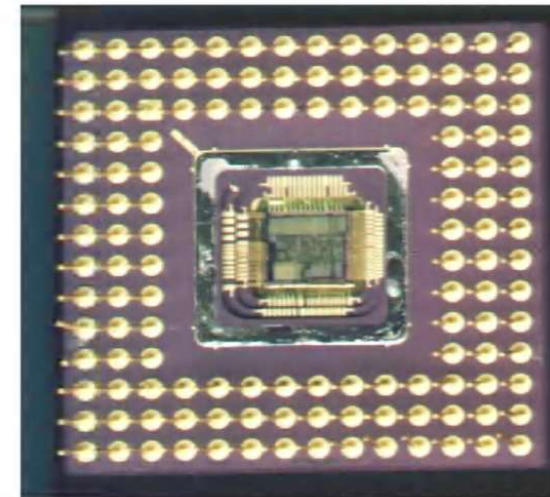
- x86 Microprocessors - 80286
 - Bus: 16-bit data bus; 24-bit address bus.
 - Maximum memory space: 16MB
 - Supports virtual memory management
 - Supports protected mode operation.
 - Protection mechanisms:
 - Segment limit checking
 - Read-only and Execute-only selection
 - 4 privilege levels.



❖ Microprocessors based on IA-32 and Intel 64 architectures.

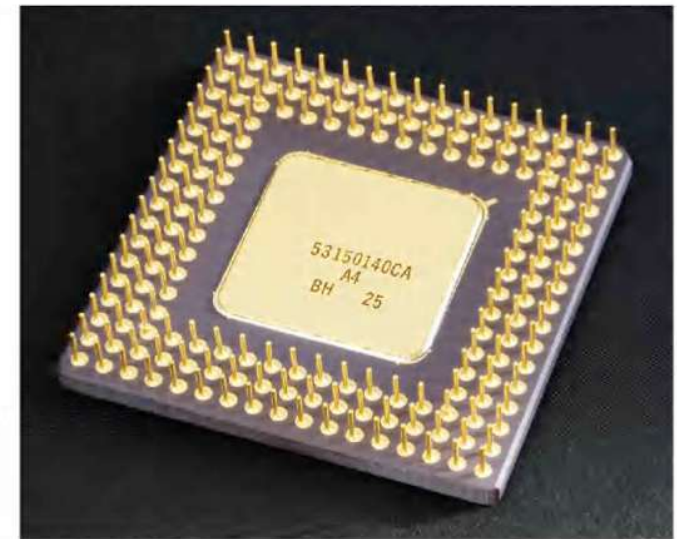
■ x86 Microprocessors - 80386

- The first 32-bit microprocessor of the IA-32 architecture
- Uses 32-bit registers
- 32-bit data bus, 32-bit address bus
- Maximum memory space: 4GB
- Good compatibility with 8086
- and 286 microprocessors
- Supports parallel instruction execution
- Supports memory management methods:
 - Segment memory model
 - Page memory model
 - Flat memory model



❖ Microprocessors based on IA-32 and Intel 64 architectures.

- x86 microprocessors - 80486
 - Better parallel instruction execution support than 386
 - Execution is divided into 5 stages:
5 instructions run in parallel in the pipeline
 - Integrated 8KB L1 cache
 - Integrated floating-point coprocessor 487
 - Integrated power management and system management capabilities.



❖ Microprocessors based on IA-32 and Intel 64 architectures.

■ Pentium I microprocessors

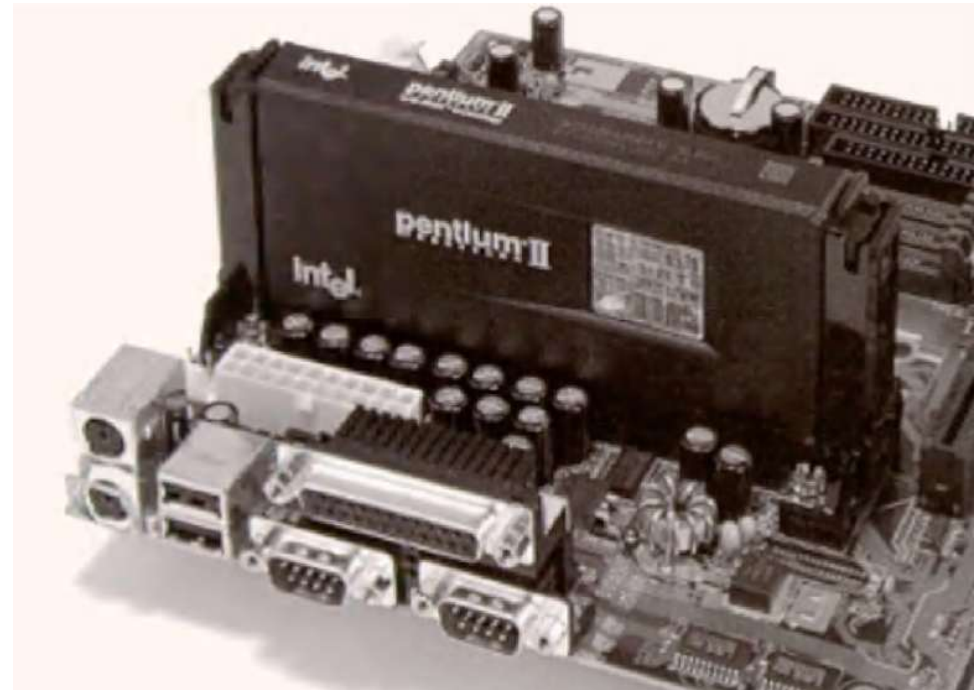
- Support two instruction pipelines:
u and v pipelines can execute 2 instructions/1 clock cycle.
- Integrated 8KB L1 cache for instruction code and 8KB L1 cache for data.
- Integrated branch prediction capability.
- 128 and 256-bit internal data lines.
- External data bus can be increased to 64 bits.
- Supports MMX technology (using SIMD).



❖ Microprocessors based on IA-32 and Intel 64 architectures.

■ Pentium II Processors

- Support advanced MMX technology.
- Integrated 16KB L1 cache for instruction code and 16KB L1 cache for data.
- Integrated L2 cache with multiple options: 256, 512, and 1MB.
- Support for advanced power management features.
- Uses Slot 1 type socket.



❖ Microprocessors based on IA-32 and Intel 64 architectures.

■ Pentium III Processors

- SSE (Streaming SIMD Extensions) Instruction Set
 - Supports 128-bit register set
 - Accelerates 3D graphics instructions
- Operating frequency from 450MHz - 1.4GHz
- Integrated dual cache
- Uses Socket 370 slot.



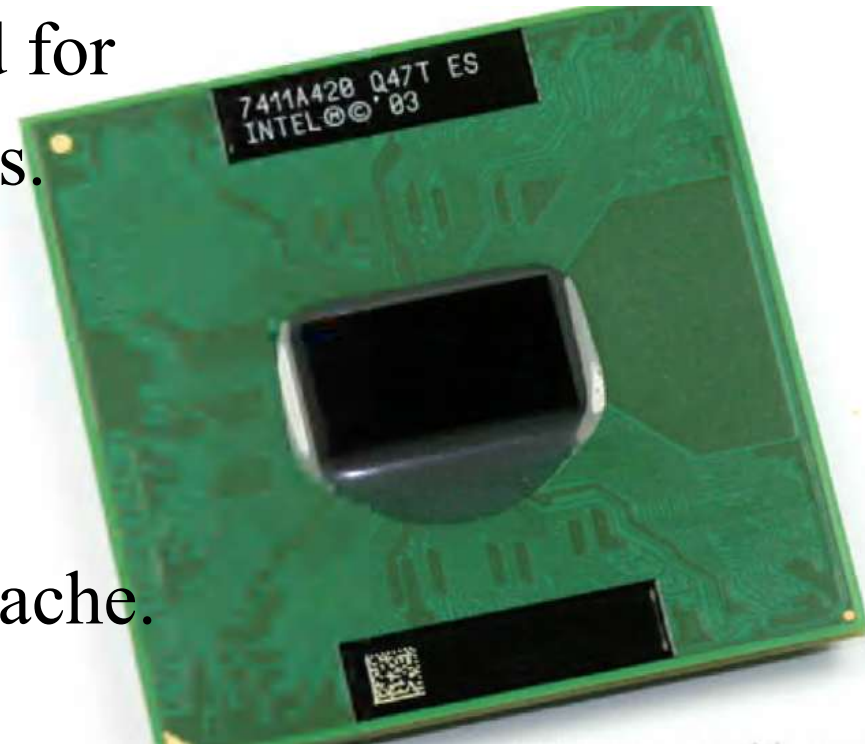
❖ Microprocessors based on IA-32 and Intel 64 architectures.

■ Pentium IV processors

- Based on the Intel NetBurst microarchitecture
- Support advanced instruction sets SSE, SSE2, SSE3.
- Some newer Pentium 4 versions support hyper-threading technology
- Pentium 4 versions 672 and 662 support virtualization technology.



- ❖ Microprocessors based on IA-32 and Intel 64 architectures.
 - Pentium M processors are designed for low-power, high-performance laptops.
 - Advanced features of Pentium M:
 - Dynamic execution
 - 32KB L1 data cache, 32KB L1 instruction cache, up to 2MB L2 cache.
 - Advanced branch prediction and data fetch logic.
 - Support for MMX, SSE, and SSE2 instruction sets.
 - Intel Speedstep advanced power management technology.



❖ Microprocessors based on IA-32 and Intel 64 architectures

■ Xeon series processors

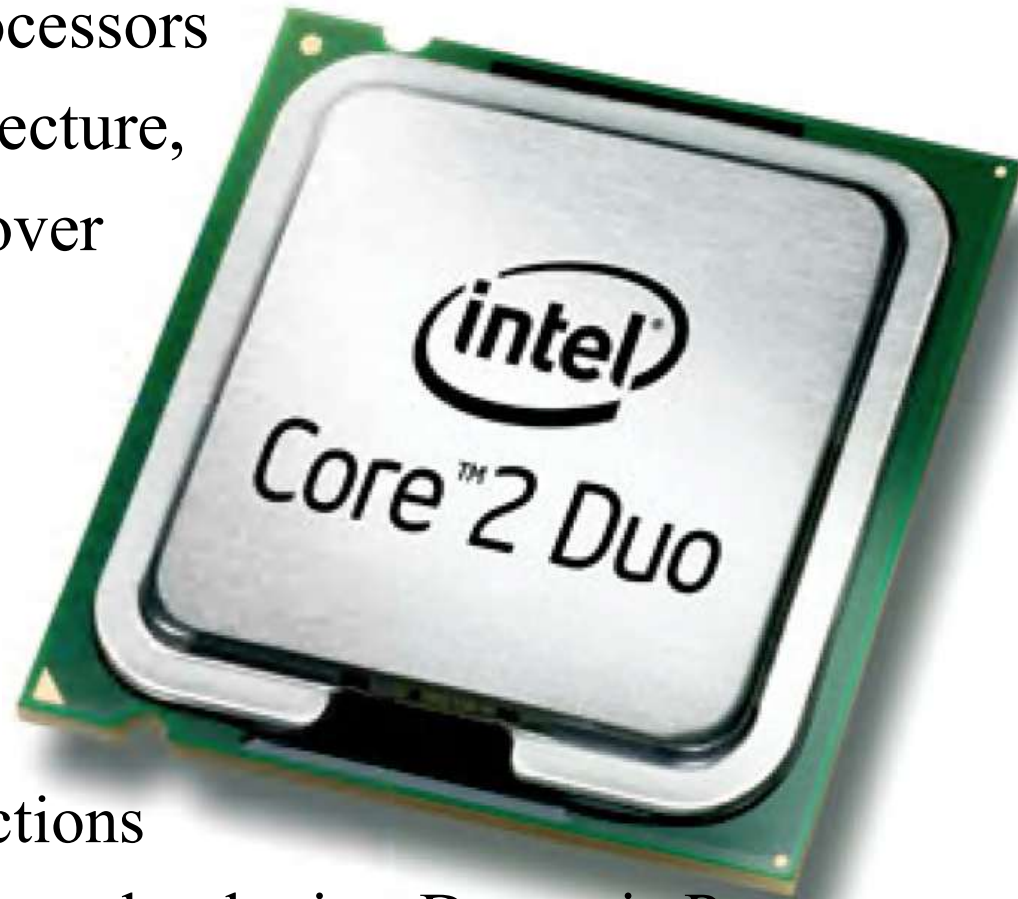
- Designed specifically for high-performance, multi-CPU server systems.
- Intel Xeon processors are based on the Intel P6, NetBurst, and Core microarchitectures.
- The Intel Xeon MP series supports hyper-threading technology.
- The Intel Xeon 5100 series is based on the energy-efficient and high-performance Core and Intel 64 microarchitectures. It also supports virtualization technology.



❖ Microprocessors based on IA-32 and Intel 64 architectures

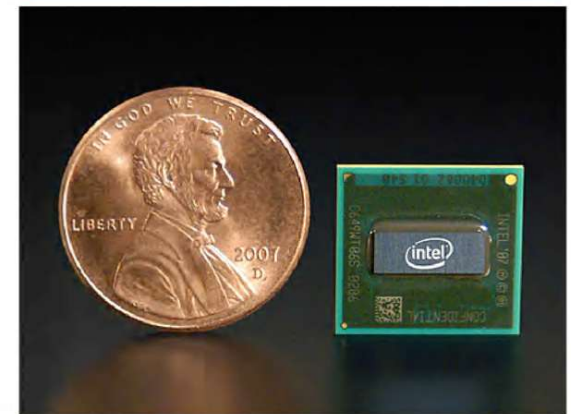
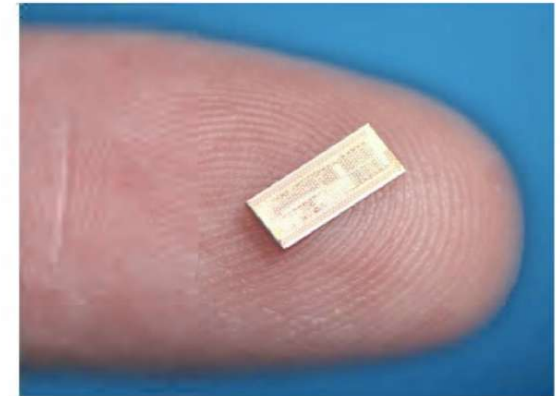
■ Core Duo & Core Solo series processors

- Based on the Core microarchitecture, offering several improvements over the Pentium M.
- Advanced features:
 - Smart cache allows data sharing between two cores
 - Improved phase decoding and execution of SIMD instructions
 - Power consumption reduction technologies: Dynamic Power Coordination and Enhanced Intel Deep Sleep.
 - Intel Advanced Thermal Manager uses digital sensors.



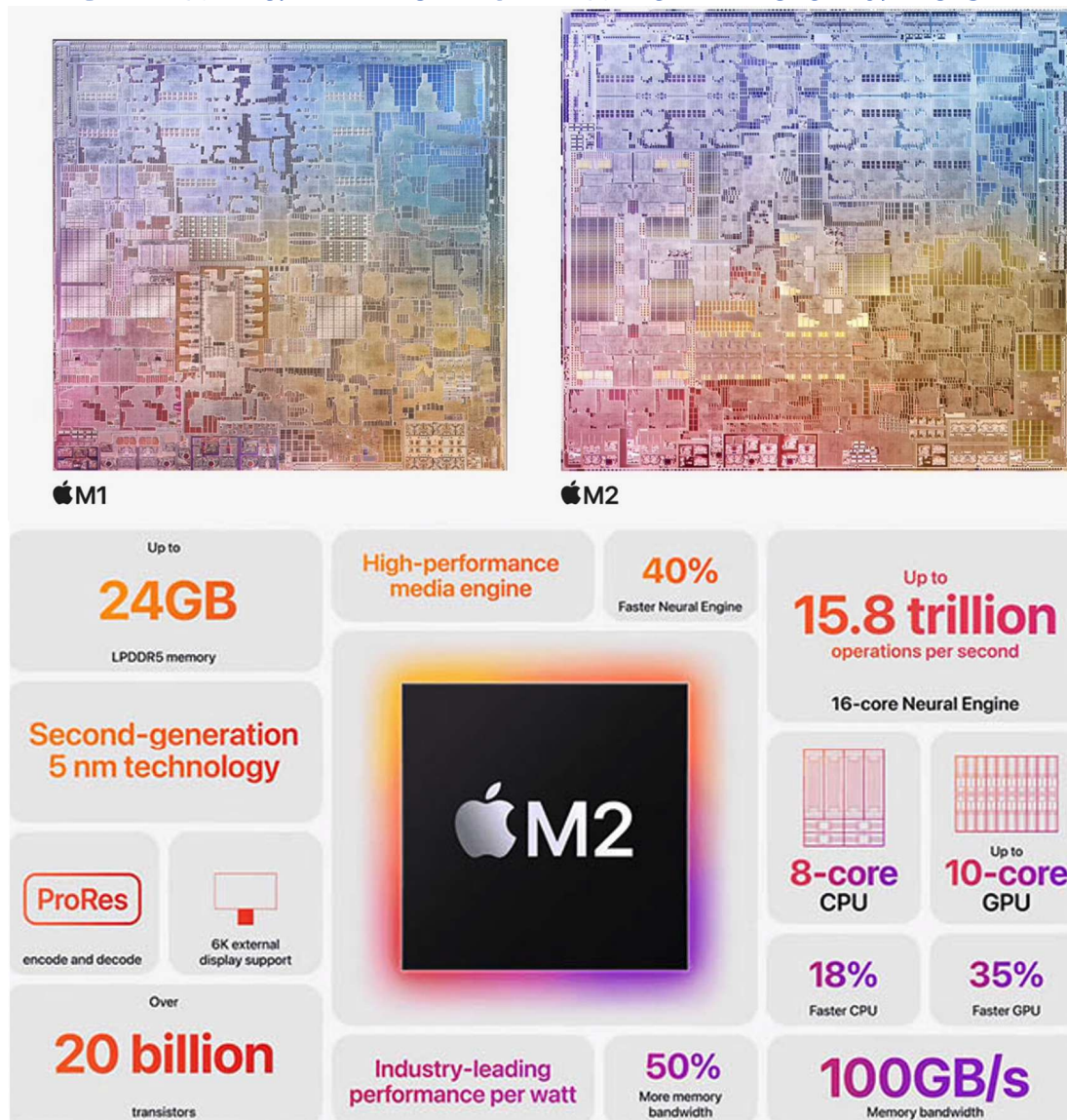
❖ Microprocessors based on IA-32 and Intel 64 architectures

- Atom series processors
 - Based on the Atom microarchitecture and 45nm technology
 - The Atom microarchitecture is optimized for small-sized devices and low power consumption.
 - Advanced features:
 - SpeedStep Boost technology
 - Deep power-off technology with dynamic cache size
 - Intel virtualization technology
 - Intel 64 architecture



❖ Microprocessors based on IA-32 and Intel 64 architectures

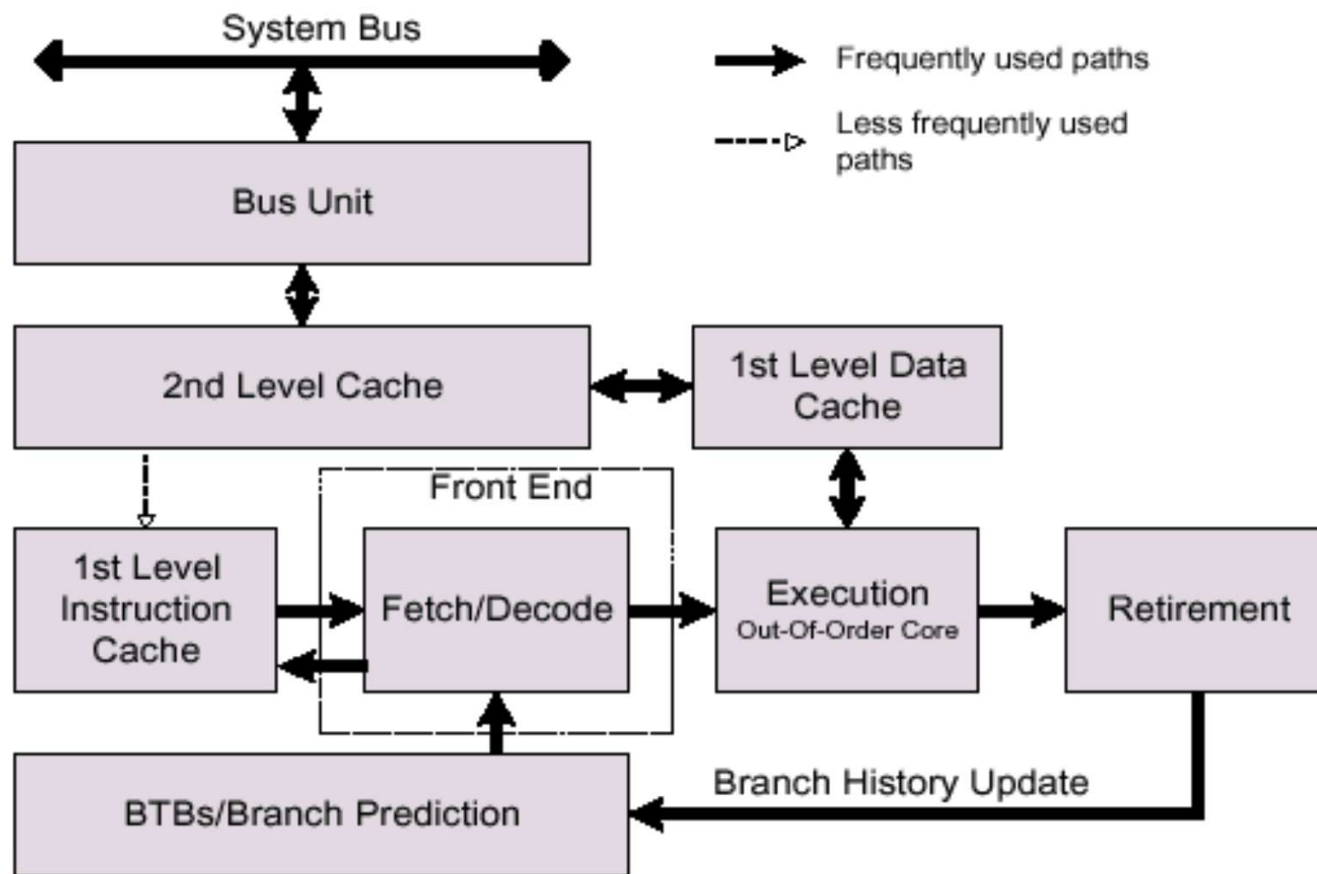
- Architecture of M1 and M2 microprocessors



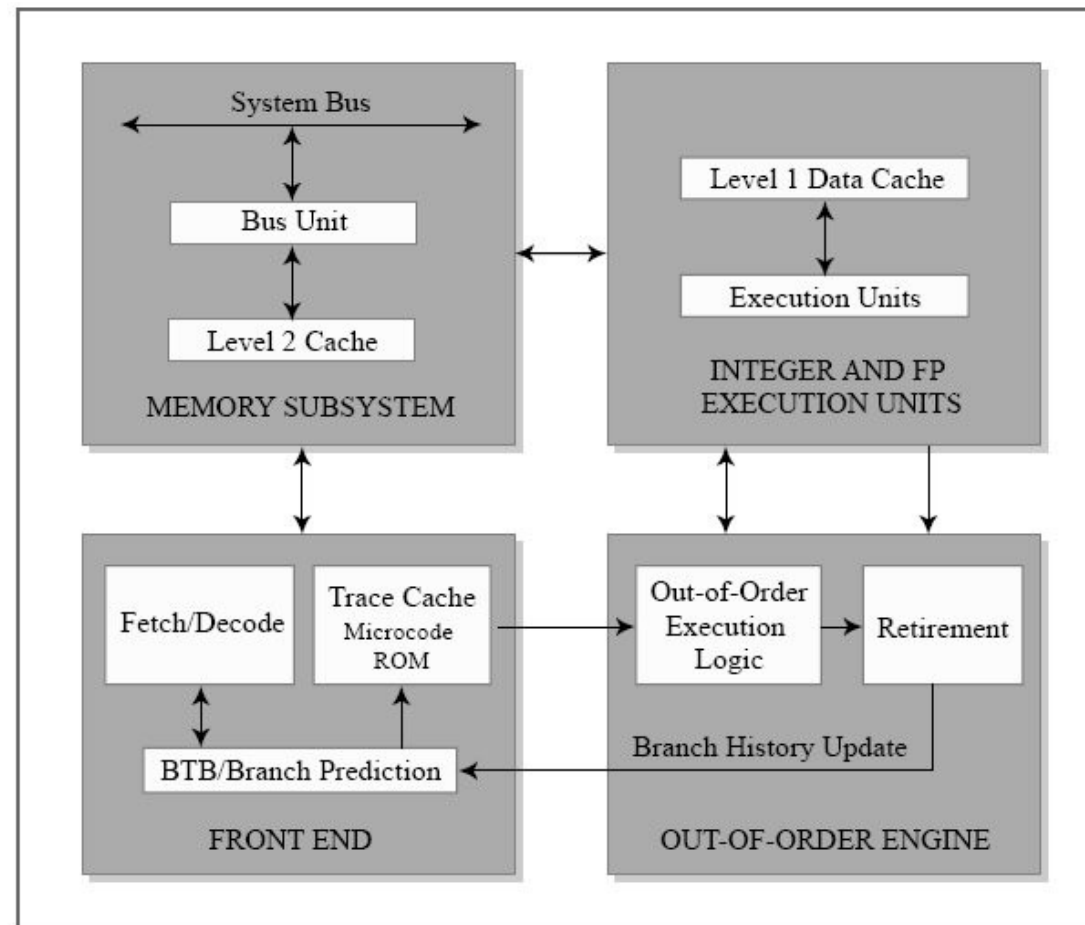
- ❖ Some advanced microprocessor architectures and technologies:
 - P6 microarchitecture
 - Intel NetBurst microarchitecture
 - Intel Core microarchitecture
 - Atom microarchitecture
 - Intel 64 architecture
 - Virtualization technology

❖ Some advanced microprocessor architectures and technologies:

■ P6 microarchitecture

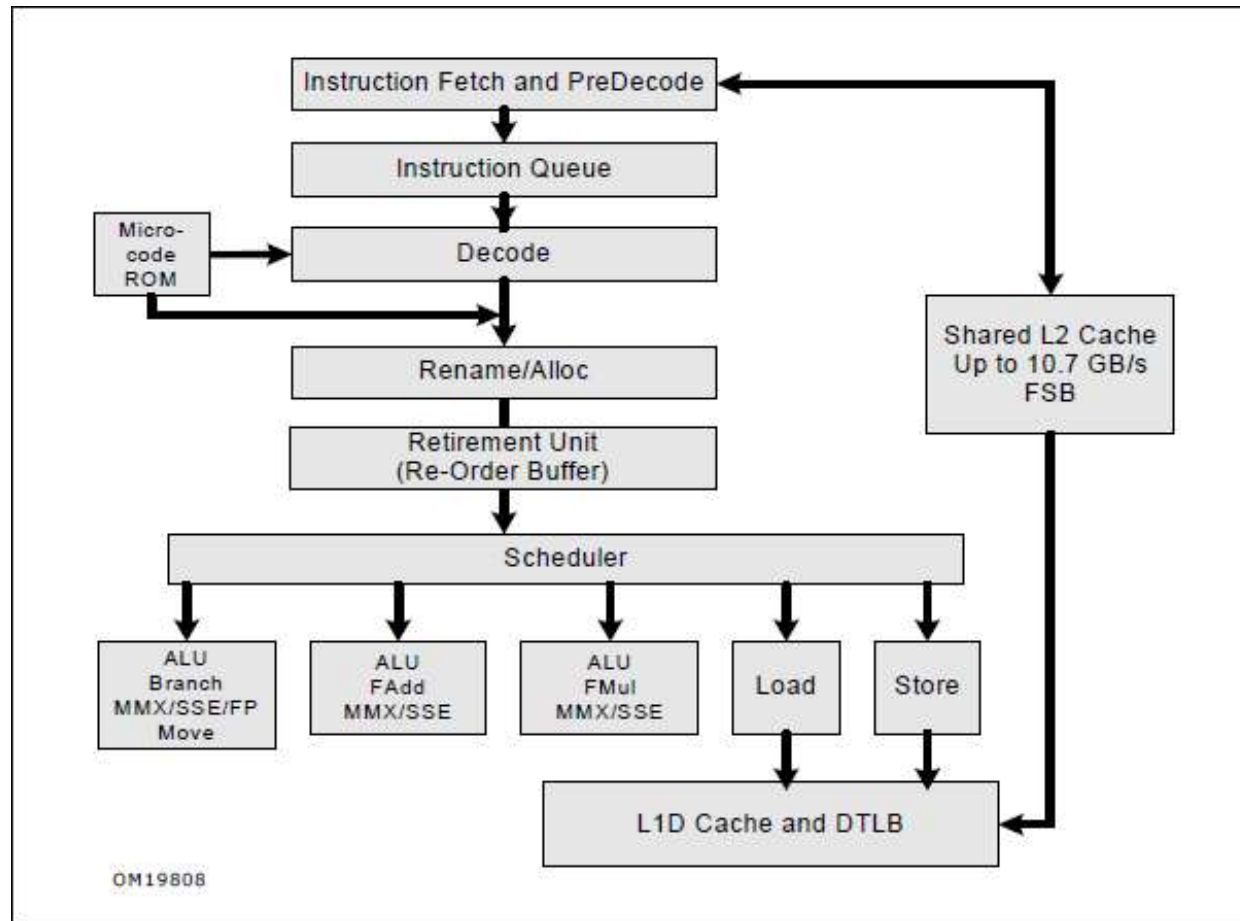


- ❖ Some advanced microprocessor architectures and technologies:
 - Intel NetBurst microarchitecture

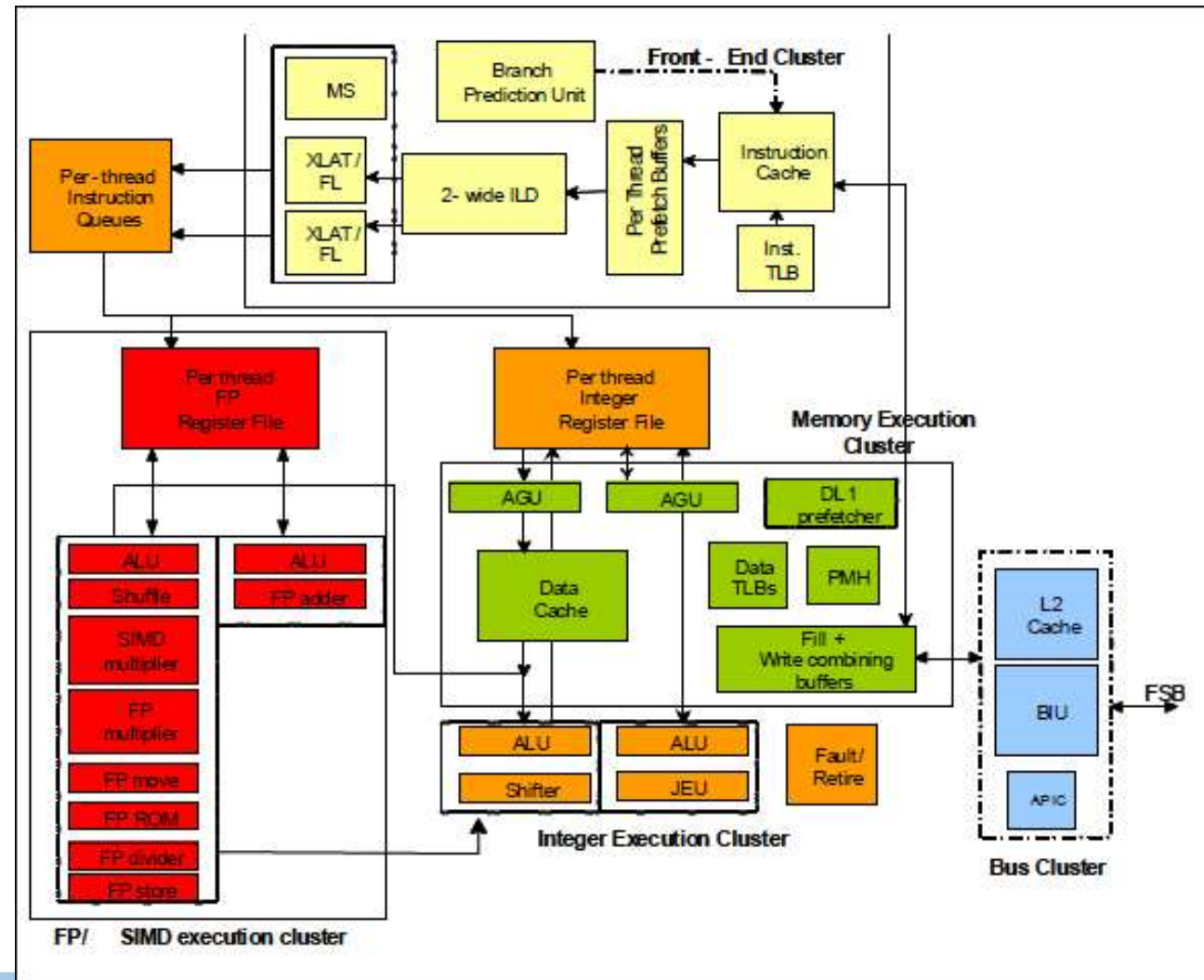


❖ Some advanced microprocessor architectures and technologies:

■ Intel Core microarchitecture

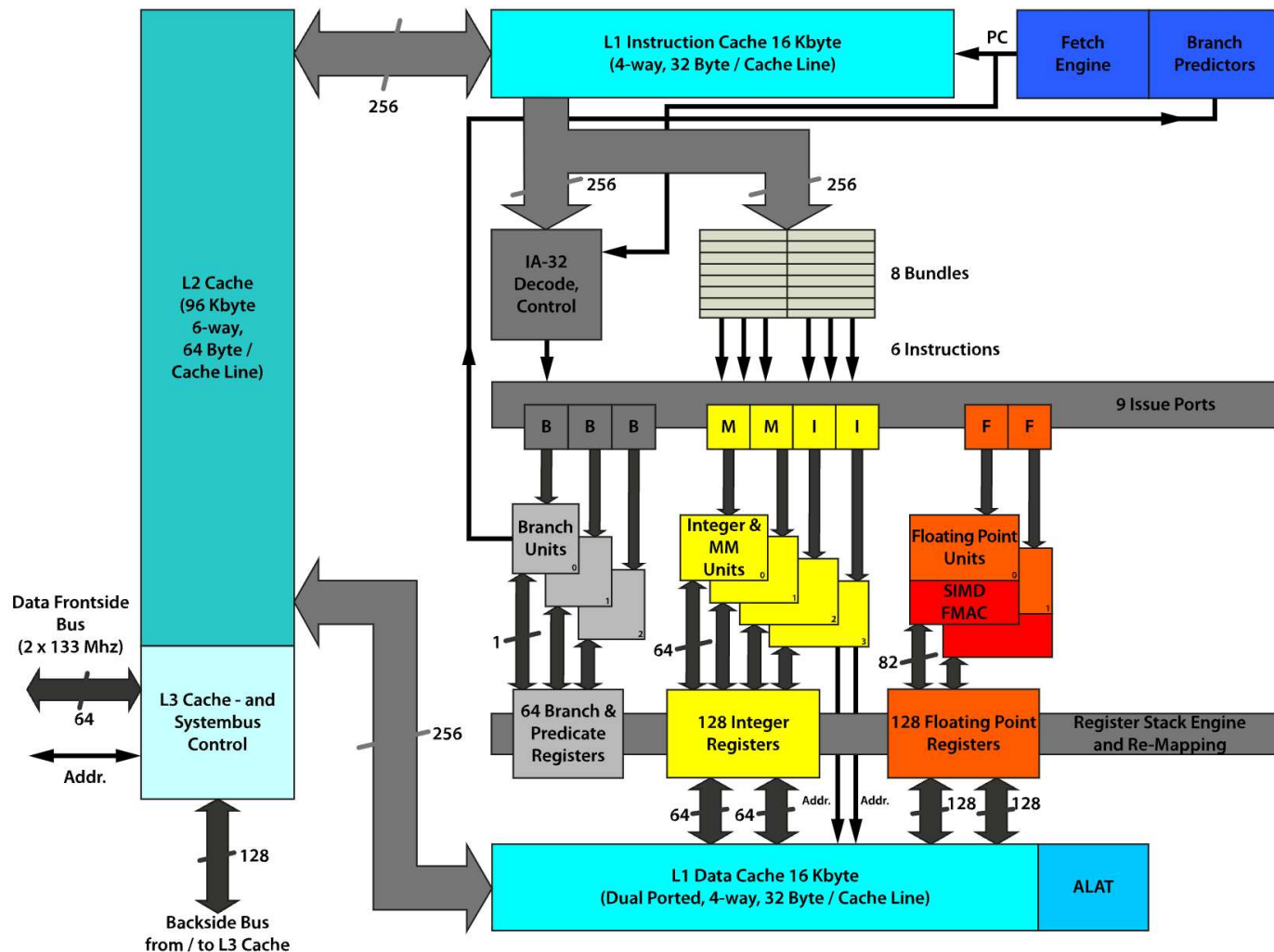


- ❖ Some advanced microprocessor architectures and technologies:
 - Atom microarchitecture



❖ Some advanced microprocessor architectures and technologies:

■ Intel 64 architecture

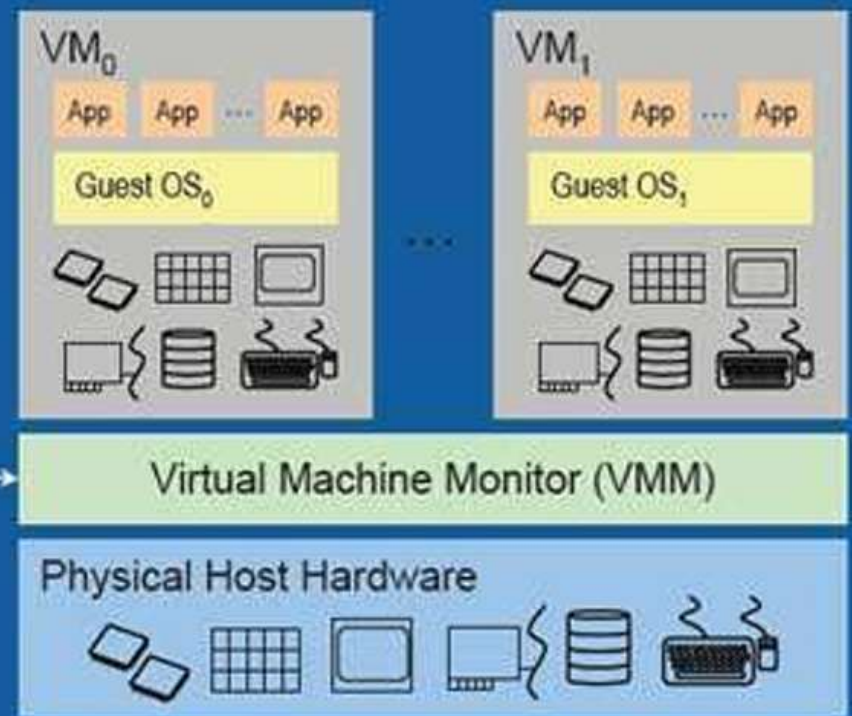


- ❖ Some advanced microprocessor architectures and technologies:
 - Virtualization technology



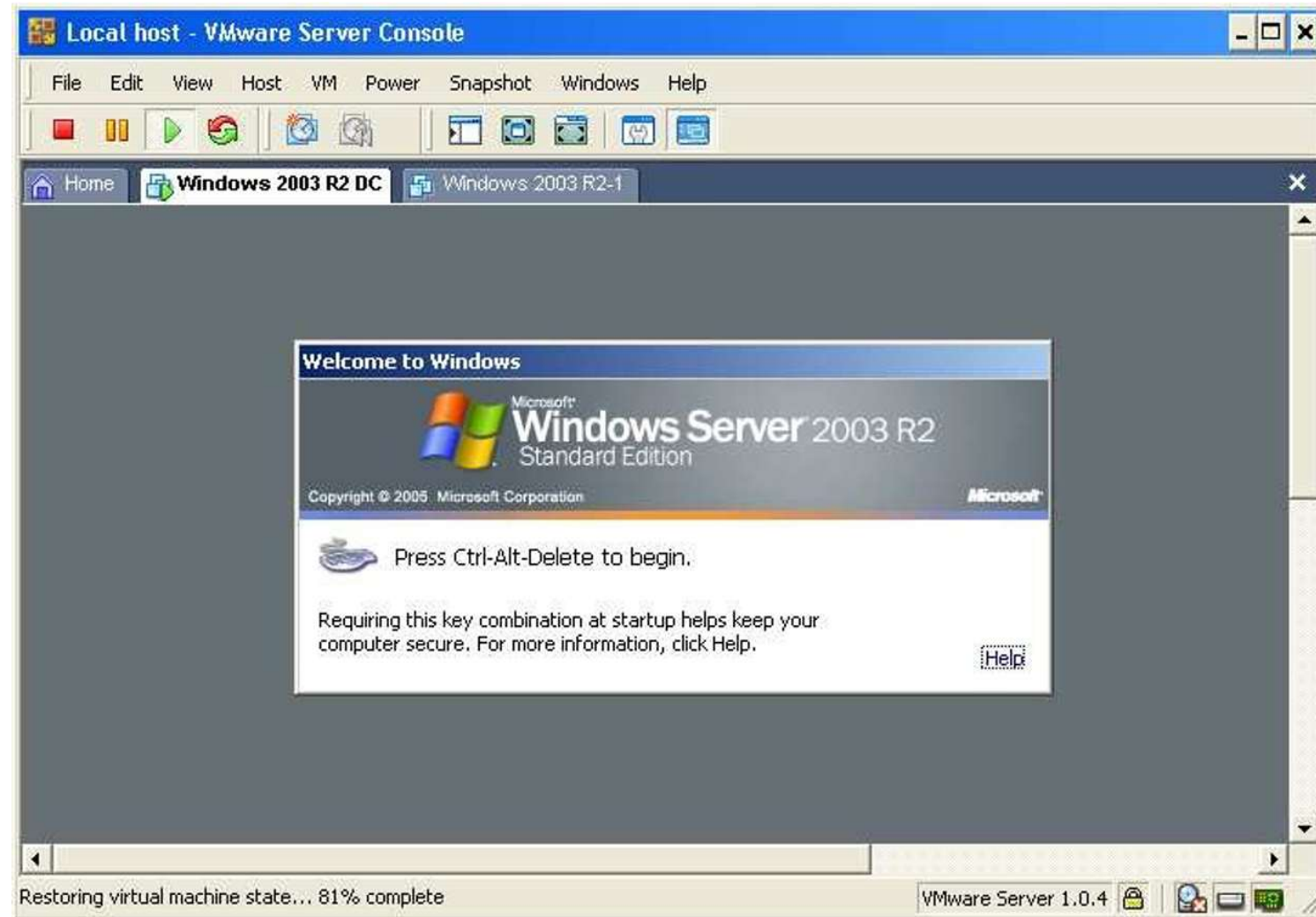
Without VMs: Single OS owns all hardware resources

A new layer of software...



With VMs: Multiple OSes share hardware resources

- ❖ Some advanced microprocessor architectures and technologies:
 - Virtualization technology – VMWare



- ❖ Question 1: What are the roles and methods of data input/output? What are the different data input/output methods?
- ❖ Question 2: What are polling input/output, interrupt input/output, and direct memory access (DMA) input/output? Compare the similarities and differences between these methods and provide illustrative examples for each.